## CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The mean of 6 numbers is 14. If every number is multiplied by 5 and then 3 is added, the new mean is: 1

**(a)** 73 **(b)** 17 **(c)** 70 **(d)** 85

**2.** The mean of 5 numbers is 18. When one number is removed the mean of the remaining 4 becomes 20. The removed number is: 1

**(a)** 10 **(b)** 28 **(c)** -2 **(d)** 18

**3.** The mean of 12 values was found to be 50. Later it was found that 62 was misread as 26. The correct mean is: 1

**(a)** 86 **(b)** 53 **(c)** 50 **(d)** 47

**4.** A class has 19 students with mean age 11 years. When the teacher's age is included, the mean rises to 12. The teacher's age (years) is: 1

**(a)** 12 **(b)** 31 **(c)** 26 **(d)** 30

**5.** The mean of the first 30 odd natural numbers is: 1

**(a)** 15.5 **(b)** 31 **(c)** 900 **(d)** 30

**6.** A data set 20, 19, 32, 56, 22, 34, 29 gets an extra (outlier) value 200. The median of the new data of 8 values is: 1

**(a)** 51.5 **(b)** 30.5 **(c)** 29 **(d)** 200

**7.** A car travels 150 km at 50 km/h and then 150 km at 75 km/h. Its average speed (km/h) for the whole journey is: 1

**(a)** 30 **(b)** 115 **(c)** 62.5 **(d)** 60

**8.** The mean of 12 numbers is 9. If every number is multiplied by 4 and then 10 is added, the new mean is: 1

**(a)** 46 **(b)** 76 **(c)** 36 **(d)** 19

**9.** **Assertion (A):** The range of the data 12, 25, 7, 40, 19 is 33. 1
**Reason (R):** Range = largest value − smallest value.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** A dot plot shows how many times each value occurs in a data set. 1
**Reason (R):** The mean is the sum of the observations divided by their number.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A data set was recorded as 29, 34, 39, 40, 15, 41, but the first value should have been 23. Find the correct mean. 2

**12.** The mean of 6 numbers is 48. When one number is removed, the mean of the remaining 5 numbers is 45. Find the removed number. 2

**13.** The mean of the first 20 even natural numbers is 21. Verify this for n = 20. 2

**14.** A data set was recorded as 51, 31, 12, 23, 16, 35, but the first value should have been 45. Find the correct mean. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A class of 50 students has mean mark 41. After rechecking, a score of 30 was corrected to 60. Find the new mean. 3

**16.** The mean age of 19 players is 17 years. When the coach is included the mean becomes 19. Find the age of the coach. 3

**17.** A student's marks in 5 tests are 55, 46, 60, 47, 62. What mark must she score in the 6th test so that her mean becomes 60? 3

OR

A class of 35 students has mean mark 44. After rechecking, a score of 35 was corrected to 70. Find the new mean. 3

**18.** The mean age of 24 players is 28 years. When the coach is included the mean becomes 30. Find the age of the coach. 3

OR

A student's marks in 5 tests are 60, 50, 55, 62, 58. What mark must she score in the 6th test so that her mean becomes 63? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** The line graph of the maximum temperature (in °C) of a town in Gujarat on five days reads: Monday 28, Tuesday 30, Wednesday 33, Thursday 31, Friday 28. 5
(i) On which day was the temperature the highest?
(ii) By how much did it rise from Monday to Wednesday?
(iii) Between which two consecutive days was the fall the greatest?
(iv) Find the mean temperature and the number of days above the mean.

**20.** The mean of 10 numbers is 35, the mean of 20 numbers is 15 and the mean of 20 numbers is 70. Find the mean of all 50 numbers. 5

OR

The mean of 25 observations is 16. If each observation is multiplied by 3 and then decreased by 7, find the new mean and the new sum. 5

## END OF QUESTION PAPER — Total: 40 Marks